

Module 11: Genomics and Biotechnology — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. **Explain** the function of restriction enzymes and DNA ligase in creating recombinant DNA.
2. **Describe** the steps and purpose of PCR and gel electrophoresis.
3. **Evaluate** the benefits and risks of transgenic organisms (GMOs).
4. **Discuss** the goals and methods of gene therapy.

Key Terminology Checklist

Define these terms in your own words to ensure mastery.

- [] **Recombinant DNA:** DNA containing genetic material from two different sources.
- [] **Plasmid:** A small circular DNA molecule in bacteria, often used as a cloning vector.
- [] **DNA Ligase:** An enzyme that seals breaks in the DNA backbone.
- [] **STR (Short Tandem Repeat):** Highly variable DNA sequences used in genetic fingerprinting.
- [] **Bioinformatics:** The use of computational tools to analyze biological data.

Concept Check

1. Molecular Cloning

- **Question:** What enzymes are used to create recombinant DNA?
- **Key Answer:**
- **Restriction enzymes:** Cut DNA at specific recognition sequences, generating sticky or blunt ends.
- **DNA ligase:** Joins DNA fragments together.

2. Gel Electrophoresis

- **Question:** How does gel electrophoresis separate DNA?
- **Key Answer:** DNA is negatively charged (phosphate groups). In an electric field, DNA migrates toward the positive electrode. Smaller fragments move faster/farther through the gel matrix.

3. Transgenic Organisms

- **Question:** What are applications of transgenic technology?
- **Key Answer:**
- **Bacteria:** Production of insulin, human growth hormone.
- **Plants:** Golden Rice (Vitamin A), pest-resistant crops.
- **Animals:** Pharmaceutical production in milk.